Do Now · retrieval

Lesson 122 · Selecting a Development Method: Case Study

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What Boolean operator joins invalid cases score<0 and score>100?

Q2

A linked-list node has data and what other essential field?

Q3

In a BST, where is a value less than the current node placed?

Q4

Why can an array provide direct indexed access?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Analyse the brief — Separate stakeholder requirements, constraints, risk and available resources.

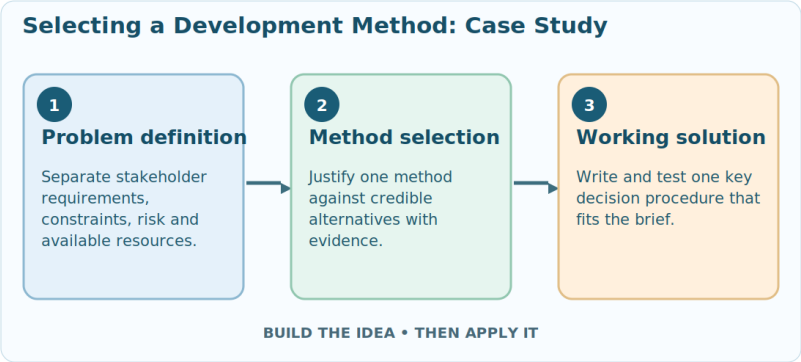
Core idea 2

Choose a lifecycle — Justify one method against credible alternatives with evidence.

Core idea 3

Trace a core algorithm — Write and test one key decision procedure that fits the brief.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

functional requirement — Behaviour a system must provide.

non-functional requirement — Quality or constraint on system operation.

acceptance criterion — Condition showing requirement has been met.

trade-off — Choice balancing competing advantages and costs.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

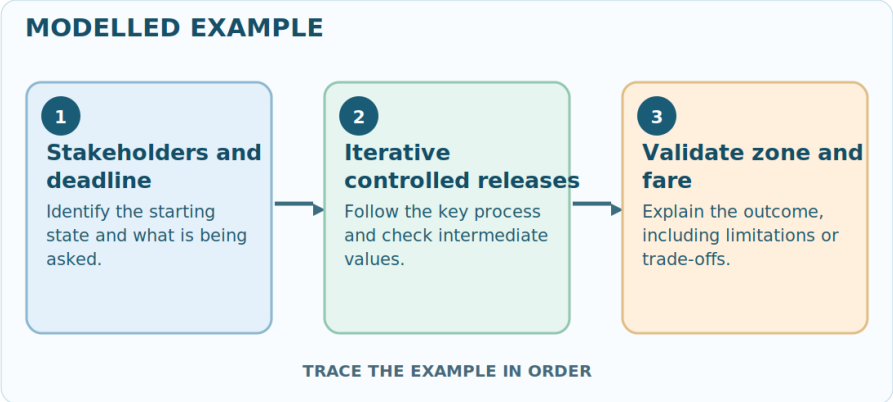
Problem

A regional bus service needs journey booking with changing fare rules and a fixed launch date. Choose an approach and an algorithm.

Follow the worked method

1. Model answer — Use iterative development for changing fare rules with documented acceptance milestones for launch. Fare algorithm: collect zones and passenger category, select valid tariff, compute fare, reject invalid input and test boundary zones. — Check: link each claim to the scenario and use precise OCR terminology.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

State two functional and two non-functional requirements for an event-booking service. [6]

Question 2

Justify a suitable lifecycle for a well-understood regulated payroll update. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Design an algorithm for rejecting bookings when capacity is full and explain test cases and lifecycle fit. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

The development method should be chosen without reference to users, change risk or deadlines.

Use key vocabulary from memory

1. Explain functional requirement in your own words.

2. Explain non-functional requirement in your own words.

3. Explain acceptance criterion in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.